

Camera calibration checkerboard -- 9 x 6 INNER CORNERS, 25.0 mm squares

board 250.0 x 175.0 mm (10 x 7 squares) | inner-corner span 200.0 x 125.0 mm | page 2 is the board

1. PRINT SETTINGS. Scale 100% / 'Actual size'. NEVER 'fit to page' or 'shrink oversized pages'.

Landscape. Duplex OFF. MATTE plain paper -- gloss reflects the light you are calibrating under.

2. VERIFY THE PRINTER on the bars below BEFORE you mount the board.

200.0 mm bar and 150.0 mm bar, measured with a steel rule. Out by >1 mm -> fix the dialog and reprint.

3. MOUNT IT DEAD FLAT. Spray-mount page 2 to foamboard, MDF, or glass -- no bubbles, no bow, no curl.

After scale, FLATNESS is the dominant calibration error: a 2 mm bow across the board biases the distortion coefficients and the RMS will still look fine. Tape at the corners only is NOT enough.

4. MEASURE THE BOARD ITSELF (the board is its own ruler -- do this even if you bought a board).

Outermost inner corner to outermost inner corner: 200.0 mm along the LONG axis, 125.0 mm along the short axis.

Full black extent: 250.0 x 175.0 mm. Divide the long span by 8 to get your ACTUAL square size and pass THAT to --square. Do not assume the nominal value.

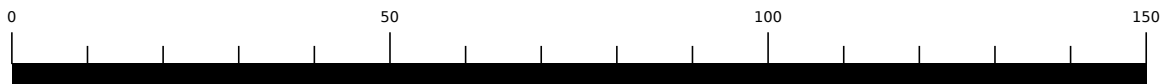
5. CAPTURE >= 15 views (build_tab skr-06 acceptance: RMS reprojection <= 1.0 px).

Vary angle, distance and position; fill the corners of the frame, not just the centre; keep the board sharp (it is the LENS you are measuring, not the paper). Shoot in the FINAL tripod geometry, and re-shoot a short set at end of session to catch thermal drift (tripod_test_protocol.md 5).

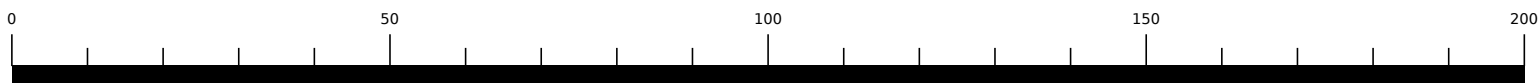
6. RUN:

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scripts/calibrate_camera.py --images 'calib/*.png' --cols 9 --rows 6 --square 0.025 --out calib.json
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Every range and bearing the campaign produces is wrong until this passes. A uniform print-scale error is INVISIBLE in the RMS -- it is consistent with itself. The bars in step 2 and the measurement in step 4 are the only things standing between a 3% print error and a 3% error in every number.



THIS BAR MUST MEASURE EXACTLY 150.0 mm



THIS BAR MUST MEASURE EXACTLY 200.0 mm

